

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA5113

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO5: *Develop the network security strategies based on the study of viruses, worms, firewall architectures, and IDS/IPS functionalities.* CO (BL5 , BL6)

Assessment Tool Weightage: 20%

QUESTIONS: 5 TOTAL MARKS: 50 Annexure A
PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)

Code of Conduct:

I DINDU PUJITHA(192373037) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: D PUJITHA

Annexure A

PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)

1. Write pseudocode to simulate a simplified TLS handshake process, including client request, server authentication using a digital certificate, session key generation, and secure data exchange.

Answer:

Simplified TLS Handshake Process

BEGIN

Client → Server: Send ClientHello with supported TLS versions and cipher suites

Server → Client: Send ServerHello with selected TLS version and cipher suite

Server → Client: Send Digital Certificate

Client:

 Verify server digital certificate

 IF certificate is invalid THEN

 Abort connection

 Display "Server authentication failed"

 STOP

 END IF

Client and Server:

 Perform key exchange

 Generate shared session key

Client → Server: Send encrypted "Finished" message

Server → Client: Send encrypted "Finished" message

IF handshake messages are verified THEN

Secure channel is established

ELSE

Abort connection

STOP

END IF

WHILE secure communication is required DO

Client encrypts application data using session key

Client → Server: Send encrypted data

Server decrypts received data

Server encrypts response using session key

Server → Client: Send encrypted response

Client decrypts received response

END WHILE

END

2. Develop pseudocode for a firewall rule engine that filters incoming packets based on IP address, port number, and protocol, and decides whether to allow or block traffic.

Answer:

Firewall Rule Engine

BEGIN

INPUT packet

Extract source_IP, destination_IP, port, and protocol

FOR each firewall_rule DO

IF packet.source_IP matches rule.source_IP

AND packet.destination_IP matches rule.destination_IP

AND packet.port matches rule.port

AND packet.protocol matches rule.protocol THEN

IF rule.action = "ALLOW" THEN

Allow packet

Log "Traffic allowed"

STOP

ELSE IF rule.action = "BLOCK" THEN

Block packet

Log "Traffic blocked"

STOP

END IF

END IF

END FOR

Block packet

Log "Traffic blocked by default rule"

END

3. Write pseudocode to detect and classify malware behavior (virus, worm, or Trojan) based on propagation method, file modification patterns, and user interaction.

Answer:

Malware Behavior Detection and Classification

BEGIN

INPUT file_or_process_behavior

propagation_method ← Analyze propagation behavior

file_modification ← Analyze file modification patterns

user_interaction ← Check whether user interaction is required

IF propagation_method = "self-replicates and infects other files"

AND file_modification = "modifies or attaches to files" THEN

malware_type ← "VIRUS"

ELSE IF propagation_method = "self-propagates across systems/networks"

AND user_interaction = "not required" THEN

malware_type ← "WORM"

ELSE IF user_interaction = "required"

AND propagation_method = "disguised as legitimate software"

AND file_modification = "may perform unauthorized changes" THEN

malware_type ← "TROJAN"

ELSE

malware_type ← "UNKNOWN OR SUSPICIOUS"

END IF

OUTPUT malware_type

Log detected behavior

END

4. Design pseudocode for an Intrusion Detection System (IDS) that uses both signature-based and anomaly-based detection to identify suspicious network activity.

Answer:

Intrusion Detection System (IDS)

```
BEGIN
  INPUT network_traffic
  Load known_attack_signatures
  Establish normal_traffic_baseline
  FOR each packet or network_event IN network_traffic DO
    # Signature-based detection
    IF event matches any known_attack_signature THEN
      Generate alert
      Log event as "Signature-based intrusion"
      CONTINUE
    END IF
    # Anomaly-based detection
    deviation ← Compare event with normal_traffic_baseline
    IF deviation > anomaly_threshold THEN
      Generate alert
      Log event as "Anomaly-based intrusion"
    ELSE
      Log event as "Normal traffic"
    END IF
  END FOR

  OUTPUT all alerts and logs
END
```

5. Create pseudocode for an Intrusion Prevention System (IPS) that monitors network traffic in real time and automatically blocks malicious requests while logging intrusion attempts.

Answer:

Intrusion Prevention System (IPS)

BEGIN

 Start real-time network monitoring

 Load attack_signatures

 Set normal_traffic_baseline

WHILE network traffic is available DO

 Capture incoming request

 Analyze request

 IF request matches known_attack_signature THEN

 Block request

 Log intrusion attempt

 Generate security alert

 ELSE

 deviation $\leftarrow$ Compare request with normal_traffic_baseline

 IF deviation > anomaly_threshold THEN

 Block request

 Log intrusion attempt

 Generate security alert

 ELSE

```
        Allow request
        Forward request to destination
    END IF
END IF
END WHILE

Stop monitoring when system is shut down
END
```